

Akhil Kota

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EDUCATION

Purdue University

West Lafayette, IN

B.S. Computer Science and Data Science (double major); GPA 4.0/4.0

Aug 2023 – May 2027

- Teaching Assistant for CS 240, 6x Dean's List, 6x Semester Honors, Outstanding Student Award Nominee
- OUR Scholarship, OUR Distinction List, SURF Distinction List, USB Best Mentor Award

EXPERIENCE

Machine Learning Researcher

Feb 2026 – Present

Li Group, Prof. Can Li (Purdue ChemE)

West Lafayette, IN

- Engineered a novel neural network architecture utilizing difference-of-convex (DC) programming to solve complex nonconvex optimization problems, ensuring high-accuracy predictive projections.
- Optimized the end-to-end training pipeline in PyTorch, implementing custom autograd workflows, feasibility-penalized loss functions, checkpointing and comprehensive metric logging
- Led the design and benchmarking of 3 novel anchor-selection algorithms, culminating in a framework slated for submission to top-tier AI conferences (ICML/ICLR 2027)

Applied ML Researcher

Sep 2025 – May 2026

Prabhakar Group, Purdue University

West Lafayette, IN

- Built an automated computer vision pipeline, integrating Cellpose and TrackMate for deep-learning segmentation and tracking across time-lapse microscopy frames; earned OUR Distinction List honors and an AAAR 44th Annual Conference submission
- Fine-tuned deep learning models on custom-annotated datasets, reducing false positive detections of microplastic debris by up to 21%
- Developed Python-based data visualizations (Matplotlib, Seaborn) to prove cytotoxicity

Software Engineer

Sep 2025 – Feb 2026

Flow

Remote

- Implemented secure, user-friendly authentication and onboarding workflows (Firebase, Google OAuth) that successfully streamlined the new user journey and boosted community engagement by 35%
- Designed a predictive driver-success model (Scikit-learn) that synthesizes location, time, and historical earnings data to deliver actionable insights.

Founding Engineer

Jan 2025 – Dec 2025

Mutually LLC

Bloomington, IN

- Built a cross-platform community based social-events platform (React Native, Expo and Supabase) and launched to 1000+ beta users on Apple TestFlight
- Designed dual encoder recommendation system (using Keras) two-tower model, lifting weekly active usage by 40% and RSVP-to-attendance conversion from 20% to 35%
- Built Selenium scraper pipeline to scrape, post-process local events and auto-publish on the platform for easy discovery

Data Science Researcher

Jan 2024 – May 2024

Cisco

West Lafayette, IN

- Optimized forecast accuracy across 15 product lines to 90% (XGBoost and Random Forest) tracking demand, pricing, and seasonality trends by product ID
- Built Plotly and Dash executive dashboards that cut manual reporting time an estimated 5 hours a week

Web Development Intern

Jun 2022 – Aug 2022

Leadzen.ai

Mumbai, India

- Interned with the Webdev team, participating in the redesign (Elementor, JetEngine) of their corporate website. Implemented custom JS, CSS to modernize the pricing module.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, HTML, CSS

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Cellpose, CVXPY, Pandas, NumPy, OpenCV

Frameworks and APIs: React, React Native, Node.js, Express.js, Flask, REST APIs, OpenAI API

Databases and Cloud: PostgreSQL, Supabase, Firebase, Google Cloud Platform

Developer Tools: Docker, GitHub Actions, Matplotlib, Seaborn, Plotly, Dash, Streamlit